Assignment 5

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Metal Recycling Program

Contract

Inputs:

- Metal ore (Mt)

Parameters:

- Day of the week
- Month
- Weekday cost of electricity ($/kWh)
- Weekend cost of electricity ($/kWh)
- Production rate (%)
- Collection rate (%)
- Recycling rate (%)
- Metal type

Outputs:

- Produced metal (Mt)
- Scrap metal (Mt)
- Recycled metal (Mt)
- Waste (Mt)

Setup

Clear the environment

▶ Show code

Load packages: here, knitr

▶ Show code

```
here() starts at /Users/nate/Documents/BREN/ESM262/Assignments/HW_5_Nathan_Sydney
```

▶ Show code

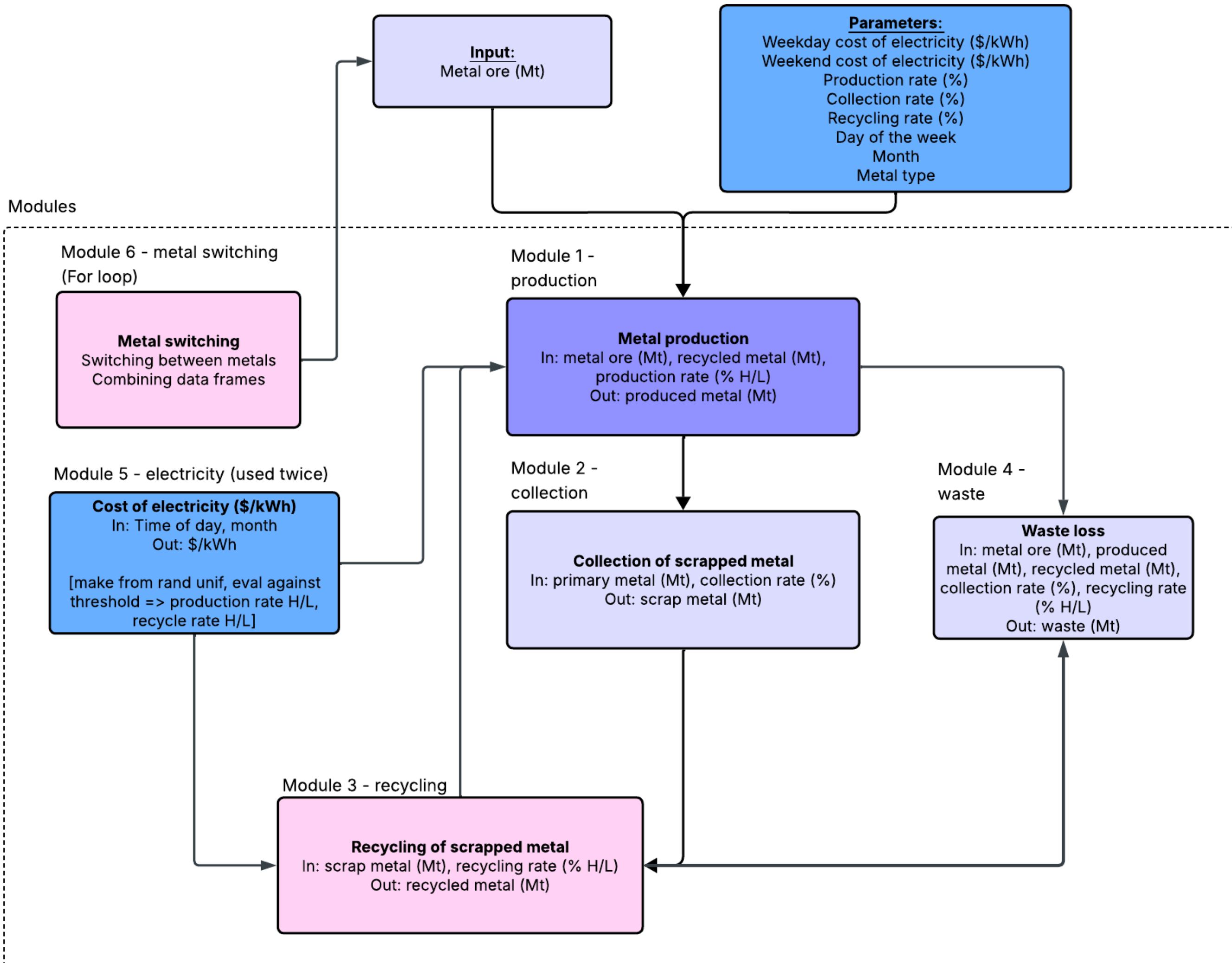
```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.2.1     ✔ readr     2.2.0
✔ forcats   1.0.1     ✔ stringr   1.6.0
✔ ggplot2   4.0.3     ✔ tibble    3.3.1
✔ lubridate 1.9.5     ✔ tidyr     1.3.2
✔ purrr     1.2.2

── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to
become errors
```

Program Schematic

▶ Show code

Metal Recycling Program Schematic



Github repo: https://github.com/sroyal-jpg/HW_5_Nathan_Sydney.git

Modules

Module 1: Cost of Electricity

For electricity cost to influence the production and recycling rates of metal, we created a random uniform distribution with electricity costs that then ranks them as high or low depending on the value of the output.

- Range 10-30 cents/kWh (based on average electricity costs)
- The function determines high vs low and reports that value and score

- High = equal to or above 16 cents/kWh
- Low = below 16 cents/kWh

▶ Show code

Testing Options

- Inputs for the cost of electric —> should be month, day
  - month input should be greater than 0 less than or equal to 12 to account for each month
    - expect_true (month > 0)
    - expect_true (month < 12)
  - day input should be the names of the days of the week, and should match the strings exactly case-wise, with capital.
    - how do i make sure it matches exactly to a range of different strings?

Run the Electricity module 2x - choose weekend or weekday

Weekend

▶ Show code

```
       day month electricity_cost electricity
1 Saturday     6          13.8425         low
```

Weekday

▶ Show code

```
      day month electricity_cost electricity
1 Tuesday     1         17.08652        high
```

Metal Ore Data frame

We created a data frame with initial metal, mass, collection rate, production, rate, and recycling rate data.

Production rate and recycling rate both use the output from the cost of electricity module.

▶ Show code

```
     metal     mass collection_rate weekend_production_rate
1    steel 39.99528            0.95                0.719461
2 aluminum 91.48716            0.70                0.719461
3   copper 50.21624            0.50                0.719461
  weekend_recycling_rate week_production_rate week_recycling_rate
1              0.7728019           0.04580768            0.412494
2              0.7728019           0.04580768            0.412494
3              0.7728019           0.04580768            0.412494
```

▶ Show code

```
      day month electricity_cost electricity
1 Tuesday     1         17.08652        high
```

▶ Show code

```
       day month electricity_cost electricity
1 Saturday     6          13.8425         low
```

Initialize Recycled Metal Value

Initialize recycled metal so that metal production eventually knows what to add once it goes through the next iterations.

▶ Show code

Module 2: Metal Production

From starting metal mass plus recycled metal, use production rate to calculate how much metal is produced.

▶ Show code

```
     metal week_produced_metal weekend_produced_metal
1    steel            1.832091               28.77505
2 aluminum            4.190814               65.82145
3   copper            2.300289               36.12863
```

Testing Options

- Things that can be changed= mass, recycled metal amount, and production rate
  - the production rate needs to be less than 1 and positive
  - amount of metal ore and recycled metal should be positive
    - expect_true (metal_ore$mass > 0)
    - expect_true (recycled_metal > 0)
    - expect_true (production_rate < 1)

Module 3: Collection

From produced metal, use collection rate to calculate how much scrap metal is collected.

▶ Show code

```
     metal week_collected_scrap weekend_collected_scrap
1    steel             1.740486                27.33629
```

```
2 aluminum          2.933570            46.07501
3   copper          1.150145            18.06432
```

Testing Options

- These values should be smaller than the production values
  - expect_true (metal_ore$collected_scrap < metal_ore$produced_metal)

Module 4: Recycling

From collected scrap metal, use recycling rate to calculate how much metal is recycled.

▶ Show code

```
     metal week_recycled_metal weekend_recycled_metal
1    steel           0.7179401               21.12554
2 aluminum           1.2100799               35.60686
3   copper           0.4744278               13.96014
```

Module 5: Waste

From the remaining produced metal not collected and collected scrap not recycled, use collection rate and recycling rate to calculate how much metal and ore waste is generated.

▶ Show code

```
     metal week_waste weekend_waste
1    steel   38.67659      12.82210
2 aluminum   89.26452      45.68707
3   copper   49.34483      32.25972
```

Module 6: Metal Switching Module

Run each module for each metal and combine the results with the original data frame.

▶ Show code

Results

Metals produced, collected, recycled, and waste generated after one full cycle.

▶ Show code

```
     metal     mass collection_rate week_production_rate week_recycling_rate
1 aluminum 91.48716            0.70           0.04580768            0.412494
2   copper 50.21624            0.50           0.04580768            0.412494
```

```
3    steel 39.99528           0.95          0.04580768           0.412494
  week_produced_metal week_collected_scrap week_recycled_metal week_waste
1            4.190814             2.933570           1.2100799   89.26452
2            2.300289             1.150145           0.4744278   49.34483
3            1.832091             1.740486           0.7179401   38.67659
```

▶ Show code

```
     metal     mass collection_rate weekend_production_rate
1 aluminum 91.48716            0.70                0.719461
2   copper 50.21624            0.50                0.719461
3    steel 39.99528            0.95                0.719461
  weekend_recycling_rate weekend_produced_metal weekend_collected_scrap
1              0.7728019               65.82145                46.07501
2              0.7728019               36.12863                18.06432
3              0.7728019               28.77505                27.33629
  weekend_recycled_metal weekend_waste
1               35.60686      53.50196
2               13.96014      35.32365
3               21.12554      17.45867
```